# MySQL Database Administration Guide

## Server Backups, Restore, User Roles and Permissions in MySQL

MySQL Database Administration (DBA) involves managing, securing, backing up, restoring, and monitoring MySQL databases used in enterprise applications. This guide is designed for: - Fresh CS/IT Engineering Graduates - SQL Corporate Training - Hands-on MySQL Learning

# Roles and Responsibilities of MySQL DBA

• Install and configure MySQL Server

• Manage database backups and recovery

• Create users and manage permissions

• Monitor database performance

• Optimize SQL queries

• Secure MySQL databases

• Perform disaster recovery

# 1. MySQL Database Backup

A database backup is a copy of MySQL data used to recover information during: - Server failure - System crash - Human error - Cyber attack - Data corruption

## Types of MySQL Backups

| Backup Type | Description |
|---|---|
| Full Backup | Complete database backup |
| Incremental Backup | Only changed data |
| Differential Backup | Changes after full backup |
| Logical Backup | Backup using mysqldump |
| Physical Backup | Copy database files |

## Create Sample Database

```
CREATE DATABASE ecommerce_db;

USE ecommerce_db;
```

## MySQL Backup Command

```
mysqldump -u root -p ecommerce_db > ecommerce_backup.sql
```

This command creates a full logical backup of the ecommerce_db database.

## Restore MySQL Backup

```
mysql -u root -p ecommerce_db < ecommerce_backup.sql
```

Used to restore the database from backup file.

### Real-Life Use Cases of Backups

• Banking systems

• E-Commerce websites

• ERP applications

• Hospital management systems

• Airline reservation systems

# 2. MySQL User Roles and Permissions

MySQL uses users and privileges to control access to databases. Permissions help: - Prevent unauthorized access - Protect sensitive data - Control operations performed by users

### Create MySQL User

```
CREATE USER 'trainer'@'localhost'
IDENTIFIED BY 'password123';
```

### Grant Permissions

```
GRANT SELECT, INSERT, UPDATE
ON ecommerce_db.*
TO 'trainer'@'localhost';
```

This user can read, insert, and update records in ecommerce_db.

### Revoke Permissions

```
REVOKE INSERT, UPDATE
ON ecommerce_db.*
FROM 'trainer'@'localhost';
```

### Apply Permission Changes

```
FLUSH PRIVILEGES;
```

Reloads MySQL permission tables.

### Common MySQL Permissions

| Permission | Purpose |
|---|---|
| SELECT | Read records |
| INSERT | Insert records |
| UPDATE | Modify records |
| DELETE | Delete records |
| CREATE | Create tables/databases |
| DROP | Delete tables/databases |
| ALL PRIVILEGES | Complete access |

# Hands-On Practice Exercises

• Create a MySQL database

• Take full backup using mysqldump
• Restore backup into another database
• Create MySQL users
• Grant SELECT permission
• Grant INSERT and UPDATE permissions
• Revoke DELETE permission
• Test user login and access

## Mini Project - Secure E-Commerce Database

• Create ecommerce_db database
• Create Customers and Orders tables
• Insert sample data
• Take database backup
• Restore backup into another database
• Create admin and employee users
• Assign permissions
• Generate security audit report

## Common MySQL DBA Mistakes

• Not taking regular backups
• Using weak passwords
• Granting ALL PRIVILEGES unnecessarily
• Ignoring failed login monitoring
• Not testing backup recovery

## MySQL DBA Interview Questions

• What is mysqldump?
• Difference between full and incremental backup?
• What is FLUSH PRIVILEGES?
• Difference between GRANT and REVOKE?
• Why are database backups important?
• How do you secure a MySQL database?

## Final Summary

| Topic | Purpose |
|---|---|
| MySQL Backup | Protect database data |
| Restore | Recover lost database |
| User Roles | Control database access |
| Permissions | Secure database operations |
| mysqldump | Create logical backups |